

TestPlatform

Rapport d'Execution de Campagne

ms_gestion2

Projet: ms_gestion

Version: 2.5

Environnement: ms_gestion_env

Debut: 20/07/2026 08:56 | Fin: 20/07/2026 08:59

Duree: 176.4 s

Rapport genere le 20/07/2026 08:59

PARTIEL

Resume executif

Total	Succes	Echecs	Erreurs	Duree
3	2	1	0	176.4 s

Taux de reussite



Evolution vs campagne precedente: UP +45.2 points

Campagne precedente #23

Corrections IA: 3 test(s) corrige(s) automatiquement, 2 avec succes

Informations du contexte

Projet	ms_gestion
Description	test

Suites

Nom	Type	Depot Git	Branche
Suite UNIT	UNIT	N/A	main
Suite INTEGRATION	INTEGRATION	N/A	main

Cas de test

Titre	Type	Priorite	Risque	Script
autha	UNIT	1	HIGH	IA
create envirenemnt	INTEGRATION	5	CRITICAL	IA
create campagne	UNIT	2	HIGH	IA

Environnement

URL Web	http://localhost:3000
URL API	N/A
Repository	https://github.com/ilyeschaabani/Automatisation_intelligente_des_tests_applicatifs
Branche	main
Base de donnees	POSTGRESQL

Resultats details

Synthese par suite

Suite	Total	Succes	Echecs	Taux
Suite UNIT	2	1	1	50%
Suite INTEGRATION	1	1	0	100%

Suite UNIT

autha

IA x3

FAILURE

Type	UNIT	Duree	12.5 s
Script	IA	Erreur	[COMPILATION_ERROR] COMPILATION ERROR :

create campagne

IA x1

SUCCESS

Type	UNIT	Duree	12.1 s
Script	IA	Erreur	[COMPILATION_ERROR] —

Suite INTEGRATION

create envirenemnt

IA x1

SUCCESS

Type	INTEGRATION	Duree	12.7 s
Script	IA	Erreur	[COMPILATION_ERROR] —

Analyse IA

autha

Erreurs extraites	<pre>[ERROR] COMPILATION ERROR : [ERROR] /C:/tmp/ms-execution/exec- 8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[13,47] cannot find symbol [ERROR] /C:/tmp/ms-execution/exec- 8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[36,13] cannot find symbol [INFO] BUILD FAILURE [ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-testCompile) on project ms_gestion: Compilation failure: Compilation failure: [ERROR] /C:/tmp/ms-execution/exec- 8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[13,47] cannot find symbol [ERROR] symbol: class SignInRequest [ERROR] location: package com.pfe.platform.ms_gestion.dto.request [ERROR] /C:/tmp/ms-execution/exec- 8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[36,13] cannot find symbol [ERROR] symbol: class SignInRequest [ERROR] location: class suites.unit.AuthControllerTest [ERROR] -> [Help 1] [ERROR] [ERROR] To see the full stack trace of the errors, re-run Maven with the -e switch. [ERROR] Re-run Maven using the -X switch to enable full debug logging. [ERROR] [ERROR] For more information about the errors and possible solutions, please read the following articles: [ERROR] [Help 1] http://cwiki.apache.org/confluence/display/MAVEN/MojoFailureException</pre>
Analyse IA	**Cause** : Le symbole <code>`SignInRequest`</code> n'est pas trouvé car la classe est censée appartenir à <code>`com.pfe.platform.authenticationmicroservice.dto`</code>, mais Maven signale qu'il la recherche dans <code>`com.pfe.platform.ms_gestion.dto.request`</code> – indication d'un packaging incorrect dans l'import ou d'un décalage entre le code source et les imports. **Correction** : Vérifier la définition réelle de la classe <code>`SignInRequest`</code> (son packaging réel), puis corriger l'import en ligne 13 : si elle est dans <code>`com.pfe.platform.authenticationmicroservice.dto.request`</code>, changer <code>`com.pfe.platform.authenticationmicroservice.dto`</code> en <code>`com.pfe.platform.authenticationmicroservice.dto.request`</code>. **Conseil** : Toujours s'assurer que les imports correspondent exactement au packaging déclaré dans les fichiers sources (souvent via un clic droit <code>*Organize Imports*</code> dans l'IDE avant compilation).

create envirenemnt

Erreurs extraites	<pre>Hibernate: drop table if exists security_vulnerabilities cascade Hibernate: drop table if exists test_cases cascade Hibernate: drop table if exists test_suites cascade Hibernate: drop table if exists users cascade Hibernate: create table campaign_testcases (execution_order integer, campaign_id bigint not null, not null, primary key (campaign_id, test_case_id)) Hibernate: create table campaigns (created_at timestamp(6), environment_id bigint not null, fin id bigint generated by default as identity, project_id bigint not null, started_at timestamp(6) varchar(255), git_branch varchar(255), name varchar(255) not null, status varchar(255) check ((('PENDING','RUNNING','FINISHED','FINISHED_WITH_ERRORS','ABORTED'))), trigger_mode varchar(255) c ('MANUAL','SCHEDULED','CI'))), primary key (id))</pre>
----------------------	---

```
Hibernate: create table compliance_results (failed_controls integer, passed_controls integer, s
total_controls integer, evaluated_at timestamp(6), id bigint generated by default as identity,
framework varchar(255) not null, version varchar(255), primary key (id))
Hibernate: create table environments (created_at timestamp(6), id bigint generated by default a
bigint not null, base_url_api varchar(255), base_url_web varchar(255), database_type varchar(25
varchar(255), git_repo_url varchar(255), name varchar(255) not null, primary key (id))
Hibernate: create table execution_results (assigned_to_user_id bigint, campaign_id bigint, id b
default as identity, test_case_id bigint, assigned_to varchar(255), error_message TEXT, status
key (id))
Hibernate: create table notifications (read boolean not null, created_at timestamp(6), id bigin
as identity, user_id bigint not null, link varchar(255), message TEXT, title varchar(255) not n
not null check ((type in
('VULN_ASSIGNED', 'VULN_RESOLVED', 'CAMPAIGN_COMPLETED', 'TEST_FAILED', 'TEST_RESOLVED', 'REPORT_REAL
primary key (id))
Hibernate: create table project_members (id bigint generated by default as identity, project_id
user_id bigint not null, primary key (id), unique (project_id, user_id))
Hibernate: create table projects (created_at timestamp(6), created_by bigint, id bigint generat
identity, description TEXT, git_default_branch varchar(255), git_repo_url varchar(255), name va
unique, status varchar(255) check ((status in ('ACTIVE', 'PAUSED', 'ARCHIVED'))), primary key (id
Hibernate: create table security_scans (coverage float(53), files_analyzed integer, lines_analy
completed_at timestamp(6), id bigint generated by default as identity, project_id bigint, start
branch varchar(255), commit_hash varchar(255), duration varchar(255), engine varchar(255) not n
varchar(255) not null, scan_type varchar(255) not null check ((scan_type in ('SAST', 'DAST', 'SCA
varchar(255) not null check ((status in ('RUNNING', 'COMPLETED', 'FAILED', 'SCHEDULED'))), target_
primary key (id))
Hibernate: create table security_vulnerabilities (line integer, assigned_to_user_id bigint, cre
bigint generated by default as identity, project_id bigint, scan_id bigint, updated_at timestamp
varchar(255), cwe_id varchar(255), description TEXT, endpoint varchar(255), evidence TEXT, file
fix_example TEXT, http_method varchar(255), owasp_category varchar(255), parameter varchar(255)
risk TEXT, severity varchar(255) not null check ((severity in ('CRITICAL', 'HIGH', 'MEDIUM', 'LOW
TEXT, source varchar(255), status varchar(255) not null check ((status in
('OPEN', 'IN_PROGRESS', 'RESOLVED', 'CLOSED', 'FALSE_POSITIVE'))), title varchar(255) not null, vul
null check ((vuln_type in ('SAST', 'DAST', 'SCA'))), primary key (id))
Hibernate: create table test_cases (active boolean, flaky boolean, generated boolean not null,
integer, priority integer, created_at timestamp(6), id bigint generated by default as identity,
null, database_type varchar(255), description TEXT, generated_code TEXT, git_repo_url varchar(2
varchar(255) check ((risk_level in ('CRITICAL', 'HIGH', 'MEDIUM', 'LOW'))), script_path varchar(25
varchar(255), tags varchar(255), target_class_name varchar(255), title varchar(255) not null, t
((type in ('WEB', 'API', 'UNIT', 'INTEGRATION', 'UX_WEB', 'UX_MOBILE'))), test_data JSONB, primary k
Hibernate: create table test_suites (created_at timestamp(6), id bigint generated by default as
bigint not null, description TEXT, git_branch varchar(255), git_repo_url varchar(255), module_p
varchar(255) not null, type varchar(255) check ((type in ('WEB', 'API', 'UNIT', 'INTEGRATION', 'UX_
primary key (id))
Hibernate: create table users (id bigint not null, email varchar(255), image_url varchar(255),
varchar(255), primary key (id))
Hibernate: alter table if exists campaign_testcases add constraint FKdyri4dtm81vciauhdyekpeh1t
(campaign_id) references campaigns
Hibernate: alter table if exists campaign_testcases add constraint FK9weg2djl986e8fnmb11onvyg6
(test_case_id) references test_cases
Hibernate: alter table if exists campaigns add constraint FKmyimryhxtshr5smm4vb6y7ny8 foreign k
references environments
Hibernate: alter table if exists campaigns add constraint FK8tqxs91vmi30cb4pkapa432mi foreign k
references projects
Hibernate: alter table if exists compliance_results add constraint FKbc45lcknv6hj2rrknhttx0rsu
references projects
Hibernate: alter table if exists environments add constraint FKnlc5pk4efot15s7hyrqah6ru4 foreig
references projects
Hibernate: alter table if exists project_members add constraint FKdki1sp2homqsdvcvm9yrix31g for
references projects
Hibernate: alter table if exists security_scans add constraint FKal1fccjhlqwsdiyfe0xgn8hag fore
references projects
Hibernate: alter table if exists security_vulnerabilities add constraint FK5dqkgk4x5b5h2qub892e
```

```
(project_id) references projects
Hibernate: alter table if exists security_vulnerabilities add constraint FK41kves1citph1miw1j7n
(scan_id) references security_scans
Hibernate: alter table if exists test_cases add constraint FKeuc59ejswhi3pn4lv997yd82 foreign k
test_suites
Hibernate: alter table if exists test_suites add constraint FK9m60ulvcghbrs28iuc98ug1vb foreign
references projects
2026-07-13T13:56:40.140+01:00 INFO 28344 --- [ms_gestion] [          main] j.LocalContainerEn
Initialized JPA EntityManagerFactory for persistence unit 'default'
2026-07-13T13:56:40.282+01:00 INFO 28344 --- [ms_gestion] [          main] o.s.d.j.r.query.Qu
Hibernate is in classpath; If applicable, HQL parser will be used.
2026-07-13T13:56:42.261+01:00 WARN 28344 --- [ms_gestion] [          main] JpaBaseConfigurati
spring.jpa.open-in-view is enabled by default. Therefore, database queries may be performed dur
Explicitly configure spring.jpa.open-in-view to disable this warning
2026-07-13T13:56:42.437+01:00 WARN 28344 --- [ms_gestion] [          main] .s.a.UserDetailsSe

Using generated security password: b1e338f2-3515-4e86-95bc-4b340646e04f

This generated password is for development use only. Your security configuration must be update
application in production.

2026-07-13T13:56:42.455+01:00 INFO 28344 --- [ms_gestion] [          main] r$InitializeUserDe
Global AuthenticationManager configured with UserDetailsService bean with name inMemoryUserDeta
2026-07-13T13:56:43.353+01:00 INFO 28344 --- [ms_gestion] [          main] s.integration.Envi
Started EnvironmentServiceIT in 17.487 seconds (process running for 18.954)
Hibernate: insert into projects (created_at,created_by,description,git_default_branch,git_repo_
(?,?,?,?,,?)
Hibernate: insert into project_members (project_id,user_id) values (?,?)
Hibernate: select pm1_0.id from project_members pm1_0 where pm1_0.project_id=? and pm1_0.user_i
only
Hibernate: select e1_0.id from environments e1_0 where e1_0.project_id=? and e1_0.name=? fetch
Hibernate: insert into environments
(base_url_api,base_url_web,created_at,database_type,git_branch,git_repo_url,name,project_id) va
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 19.56 s -- in
suites.integration.EnvironmentServiceIT
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 28.099 s
[INFO] Finished at: 2026-07-13T13:56:46+01:00
[INFO] -----

----- STDERR -----
Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes be
classpath has been appended
```

Analyse
IA

```
### 1. Ce qui est bien testé
Le test valide avec succès le *happy path* de la création d'un environnement via `EnvironmentSe
couvrant la validation des données d'entrée, la persistance en base (H2), et la bonne propagati
réponse. L'initialisation du contexte de sécurité (via `SecurityContextHolder`) et la préparati
(projet + membre) sont correctement gérées via `@BeforeMethod`.

### 2. Suggestion d'amélioration
Le test manque de vérification *spécifique* de la logique métier liée à l'autorisation (ex. : `
puisque le `ProjectMember` est créé *avant* l'appel – ce qui rend le test insensible aux erreur
permission. Il serait plus robuste d'ajouter un assert sur une *exception attendue* (ex. `Busin
l'utilisateur n'est *pas* membre, et/ou de mocker la méthode de vérification (si elle est extra
```

externe ou utilisant une logique complexe), afin d'isolation cette pré-condition.

(Optionnel : on pourrait aussi élargir aux cas limites – nom vide, branches invalides – mais il y a l'autorisation.)

create campagne

Erreurs extraites	<pre>[INFO] Scanning for projects... [INFO] [INFO] -----< com.pfe.platform:ms_gestion >----- [INFO] Building ms_gestion 0.0.1-SNAPSHOT [INFO] from pom.xml [INFO] -----[jar]----- [INFO] [INFO] --- resources:3.3.1:resources (default-resources) @ ms_gestion --- [INFO] Copying 1 resource from src\main\resources to target\classes [INFO] Copying 0 resource from src\main\resources to target\classes [INFO] [INFO] --- compiler:3.14.1:compile (default-compile) @ ms_gestion --- [INFO] Nothing to compile - all classes are up to date. [INFO] [INFO] --- resources:3.3.1:testResources (default-testResources) @ ms_gestion --- [INFO] skip non existing resourceDirectory C:\tmp\ms-execution\exec- 18394668852753624163\Backend\ms_gestion\src\test\resources [INFO] [INFO] --- compiler:3.14.1:testCompile (default-testCompile) @ ms_gestion --- [INFO] Recompiling the module because of changed source code. [INFO] Compiling 2 source files with javac [debug parameters release 17] to target\test-classes [INFO] [INFO] --- surefire:3.2.5:test (default-test) @ ms_gestion --- [INFO] Using configured provider org.apache.maven.surefire.testng.TestNGProvider [INFO] [INFO] ----- [INFO] T E S T S [INFO] ----- [INFO] Running suites.unit.CampaignServiceCreateHappyTest [INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.946 s -- in suites.unit.CampaignServiceCreateHappyTest [INFO] [INFO] Results: [INFO] [INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0 [INFO] [INFO] ----- [INFO] BUILD SUCCESS [INFO] ----- [INFO] Total time: 6.703 s [INFO] Finished at: 2026-07-20T08:59:38+01:00 [INFO] ----- ----- STDERR ----- Java HotSpot(TM) 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended</pre>
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Analyse IA	**1. Ce qui est bien testé :** Le test valide correctement le <i>*happy path*</i> de la création d'une campagne : la résolution des dépendances (Project, Environment), la vérification d'accès (<i>`checkMembership`</i>), la persistance avec mapping des champs (y compris l'enum <i>`TriggerMode`</i>), et l'absence d'effets de bord sur les repositories de TestCase inutilisés. **2. Suggestion d'amélioration :**
---------------	--

Utiliser un `**`ArgumentCaptor<Campaign>`**` sur ``campaignRepository.save`` pour vérifier l'état **exact** de l'entité persistée (relations ``project`/`environment``, statut ``DRAFT`` par défaut, ``createdAt`` non nul) plutôt que de se fier uniquement à la réponse DTO, ce qui renforce la robustesse contre les régressions de mapping interne.

Corrections automatiques par IA

L'IA a detecte des erreurs dans certains scripts generes et les a corriges automatiquement.

autha

3 correction(s)

FAILURE

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[3,62] package com.pfe.platform.authenticationmicroservice.Controller does not exist
[ERROR] /C:/tmp/ms-execution/exec-8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[4,59] package com.pfe.platform.authenticationmicroservice.Service does not exist
[ERROR] /C:/tmp/ms-execution/exec-8758300152830098989/Back
Résultat: Script corrigé (3153 chars), relance Maven...
Résultat: Encore en échec (exit=1).

=== Tentative 2/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[17,26] package javax.servlet.http does not exist
[ERROR] /C:/tmp/ms-execution/exec-8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[31,13] cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-testCompile) on proje
Résultat: Script corrigé (2954 chars), relance Maven...
Résultat: Encore en échec (exit=1).

=== Tentative 3/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[5,48] cannot find symbol
[ERROR] /C:/tmp/ms-execution/exec-8758300152830098989/Backend/ms_gestion/src/test/java/suites/unit/AuthControllerTest.java:[28,52] cannot find symbol
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-testCompile) on project ms_gestion: Compilati
Résultat: Script corrigé (2994 chars), relance Maven...
Résultat: Encore en échec (exit=1).
```

create envirenemnt

1 correction(s)

SUCCESS

```
=== Tentative 1/3 ===
Erreur détectée: [ERROR] COMPILATION ERROR :
[ERROR] /C:/tmp/ms-execution/exec-14957033593420335957/Backend/ms_gestion/src/test/java/suites/integration/EnvironmentServiceIT.java:[63,29] incompatible types: com.pfe.platform.ms_gestion.entity.ProjectMemberId cannot be converted to java.lang.Long
[INFO] BUILD FAILURE
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-testCompile) on project ms_gestion: Compilation failure
[ERROR] /C:/tmp/ms-execution/exec-1495
Résultat: Script corrigé (4760 chars), relance Maven...
Résultat: SUCCÈS après correction !
```

create campagne

1 correction(s)

SUCCESS

```
=== Tentative 1/3 ===
```

```
Erreur détectée: [ERROR] COMPILATION ERROR :  
[ERROR] /C:/tmp/ms-execution/exec-  
18394668852753624163/Backend/ms_gestion/src/test/java/suites/unit/CampaignServiceCreateHappyTest.java:[93,56  
] cannot find symbol  
[INFO] BUILD FAILURE  
[ERROR] Failed to execute goal org.apache.maven.plugins:maven-compiler-plugin:3.14.1:testCompile (default-  
testCompile) on project ms_gestion: Compilation failure  
[ERROR] /C:/tmp/ms-execution/exec-  
18394668852753624163/Backend/ms_gestion/src/test/java/suites/unit/CampaignServiceCreateHapp  
Résultat: Script corrigé (5083 chars), relance Maven...  
Résultat: SUCCÈS après correction !
```

Metriques

Top 5 des tests les plus lents

Test	Suite	Duree
create envirenemnt	Suite INTEGRATION	12.7 s
autha	Suite UNIT	12.5 s
create campagne	Suite UNIT	12.1 s

Temps d'execution par type

Type	Duree moyenne
UNIT	12.3 s
INTEGRATION	12.7 s

Tests instables

Aucun test instable detecte

Recommandations

- autha:

****Cause**** : Le symbole `SignInRequest` n'est pas trouvé car la classe est censée appartenir à `com.pfe.platform.authenticationmicroservice.dto`, mais Maven signale qu'il la recherche dans `com.pfe.platform.ms_gestion.dto.request` — indication d'un paquetage incorrect dans l'import ou d'un décalage entre le code source et les imports.

****Correction**** : Vérifier la définition réelle de la classe `SignInRequest` (son paquetage réel), puis corriger l'import en ligne 13 : si elle est dans `com.pfe.platform.authenticationmicroservice.dto.request`, changer `com.pfe.platform.authenticationmicroservice.dto` en `com.pfe.platform.authenticationmicroservice.dto.request`.

****Conseil**** : Toujours s'assurer que les imports correspondent exactement au paquetage déclaré dans les fichiers sources (souvent via un clic droit *Organize Imports* dans l'IDE avant compilation).

Annexe - Scripts generes par IA

create envirenemnt

```
package suites.integration;

import com.pfe.platform.ms_gestion.dto.request.CreateEnvironmentRequest;
import com.pfe.platform.ms_gestion.dto.response.EnvironmentResponse;
import com.pfe.platform.ms_gestion.entity.Environment;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.ProjectMember;
import com.pfe.platform.ms_gestion.entity.ProjectMemberId;
import com.pfe.platform.ms_gestion.repository.CampaignRepository;
import com.pfe.platform.ms_gestion.repository.EnvironmentRepository;
import com.pfe.platform.ms_gestion.repository.ProjectMemberRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.service.EnvironmentService;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.context.SecurityContextHolderImpl;
import org.springframework.test.annotation.Rollback;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.context.TestPropertySource;
import org.springframework.test.context.testng.AbstractTestNGSpringContextTests;
import org.springframework.transaction.annotation.Transactional;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Collections;

import static org.testng.Assert.*;

@SpringBootTest
@ActiveProfiles("test")
@Transactional
@Rollback(true)
@TestPropertySource(properties = {
    "spring.datasource.url=jdbc:h2:mem:testdb;DB_CLOSE_DELAY=-1;MODE=PostgreSQL",
    "spring.datasource.driverClassName=org.h2.Driver",
    "spring.datasource.username=sa",
    "spring.datasource.password=",
    "spring.jpa.hibernate.ddl-auto=create-drop",
    "spring.jpa.database-platform=org.hibernate.dialect.H2Dialect"
})
public class EnvironmentServiceIT extends AbstractTestNGSpringContextTests {

    @Autowired private EnvironmentService environmentService;
    @Autowired private ProjectRepository projectRepository;
    @Autowired private ProjectMemberRepository projectMemberRepository;
    @Autowired private EnvironmentRepository environmentRepository;
    @Autowired private CampaignRepository campaignRepository;

    private Long projectId;
    private Long userId = 1L;

    @BeforeMethod
    public void setUp() {
        // Insert prerequisite Project
        Project project = new Project();
        project.setName("AddProject");
        project.setDescription("Test project");
        project.setGitRepoUrl("https://github.com/org/webapp");
        project.setCreatedBy(userId);
    }
}
```

```
project = projectRepository.save(project);
projectId = project.getId();

// Insert ProjectMember to satisfy checkMembership
ProjectMember projectMember = new ProjectMember();
projectMember.setId(new ProjectMemberId(userId, projectId));
projectMember.setProject(project);
projectMemberRepository.save(projectMember);

// Set security context
SecurityContextHolder.setContext(new SecurityContextImpl(
new UsernamePasswordAuthenticationToken(userId, null, Collections.emptyList())));
}

@AfterMethod
public void tearDown() {
SecurityContextHolder.clearContext();
}

@Test
public void test_add_happy_path() {
// Given
CreateEnvironmentRequest request = new CreateEnvironmentRequest();
request.setName("AddProject");
request.setBaseUrlWeb("https://app.example.com");
request.setBaseUrlApi("https://api.example.com");
request.setGitRepoUrl("https://github.com/org/webapp");
request.setGitBranch("develop");
request.setDatabaseType("POSTGRESQL");

// When
EnvironmentResponse result = environmentService.add(projectId, request);

// Then
assertNotNull(result);
assertNotNull(result.getId());
assertEquals(result.getName(), "AddProject");
assertEquals(result.getBaseUrlWeb(), "https://app.example.com");
assertEquals(result.getBaseUrlApi(), "https://api.example.com");
assertEquals(result.getGitRepoUrl(), "https://github.com/org/webapp");
assertEquals(result.getGitBranch(), "develop");
assertEquals(result.getDatabaseType(), "POSTGRESQL");

// Verify persistence in DB
Environment savedEnv = environmentRepository.findById(result.getId()).orElse(null);
assertNotNull(savedEnv);
assertEquals(savedEnv.getName(), "AddProject");
assertEquals(savedEnv.getProject().getId(), projectId);
}
}
```

autha

```
package suites.unit;

import com.pfe.platform.authenticationmicroservice.Controller.AuthController;
import com.pfe.platform.authenticationmicroservice.Service.AuthenticationService;
import com.pfe.platform.authenticationmicroservice.dto.JwtAuthenticationResponse;
import com.pfe.platform.authenticationmicroservice.dto.RefreshTokenrequest;
import com.pfe.platform.authenticationmicroservice.dto.SignInRequest;
import com.pfe.platform.authenticationmicroservice.security.SecurityUtils;
import org.mockito.InjectMocks;
import org.mockito.Mock;
import org.mockito.MockedStatic;
import org.mockito.MockitoAnnotations;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;
```



```
import javax.servlet.http.HttpServletResponse;
import java.util.Optional;

import static org.mockito.ArgumentMatchers.any;
import static org.mockito.ArgumentMatchers.eq;
import static org.mockito.Mockito.*;
import static org.testng.Assert.*;

public class AuthControllerTest {

    @Mock
    private AuthenticationService authenticationService;

    @Mock
    private HttpServletResponse response;

    @InjectMocks
    private AuthController authController;

    private AutoCloseable mocks;
    private MockedStatic<SecurityUtils> mockedSecurity;

    @BeforeMethod
    public void setUp() {
        mocks = MockitoAnnotations.openMocks(this);
        mockedSecurity = mockStatic(SecurityUtils.class);
        mockedSecurity.when(SecurityUtils::getCurrentUserId).thenReturn(1L);
    }

    @AfterMethod
    public void tearDown() throws Exception {
        mockedSecurity.close();
        mocks.close();
    }

    @Test
    public void test_signIn_happy_path() {
        // Given
        SignInRequest request = new SignInRequest();
        request.setUsername("user.test");
        request.setPassword("SecurePass123!");

        JwtAuthenticationResponse authResponse = new JwtAuthenticationResponse();
        authResponse.setAccessToken("validAccessToken");
        authResponse.setRefreshToken("validRefreshToken");
        authResponse.setUsername("user.test");

        when(authenticationService.signIn(eq("user.test"), eq("SecurePass123!")))
            .thenReturn(authResponse);
        when(authenticationService.setAuthenticationCookies(any(HttpServletResponse.class), eq("validAccessToken"),
            eq("validRefreshToken")))
            .thenReturn(true);

        // When
        var result = authController.signIn(request, response);

        // Then
        assertNotNull(result);
        assertEquals(result.getStatusCode().is2xxSuccessful(), true);
        assertNotNull(result.getBody());
        assertEquals(result.getBody().getUsername(), "user.test");
        assertEquals(result.getBody().getAccessToken(), "validAccessToken");
        assertEquals(result.getBody().getRefreshToken(), "validRefreshToken");
        verify(authenticationService).signIn(eq("user.test"), eq("SecurePass123!"));
        verify(authenticationService).setAuthenticationCookies(eq(response), eq("validAccessToken"),
            eq("validRefreshToken"));
    }
}
```

create campagne

```
package suites.unit;

import com.pfe.platform.ms_gestion.dto.request.CreateCampaignRequest;
import com.pfe.platform.ms_gestion.dto.response.CampaignResponse;
import com.pfe.platform.ms_gestion.entity.Campaign;
import com.pfe.platform.ms_gestion.entity.CampaignTestCase;
import com.pfe.platform.ms_gestion.entity.Environment;
import com.pfe.platform.ms_gestion.entity.Project;
import com.pfe.platform.ms_gestion.entity.TestCase;
import com.pfe.platform.ms_gestion.entity.TestSuite;
import com.pfe.platform.ms_gestion.repository.CampaignRepository;
import com.pfe.platform.ms_gestion.repository.CampaignTestCaseRepository;
import com.pfe.platform.ms_gestion.repository.EnvironmentRepository;
import com.pfe.platform.ms_gestion.repository.ProjectRepository;
import com.pfe.platform.ms_gestion.repository.TestCaseRepository;
import com.pfe.platform.ms_gestion.security.SecurityUtils;
import com.pfe.platform.ms_gestion.service.CampaignService;
import com.pfe.platform.ms_gestion.service.ProjectAccessService;

import org.mockito.InjectMocks;
import org.mockito.Mock;
import org.mockito.MockedStatic;
import org.mockito.MockitoAnnotations;
import org.testng.annotations.AfterMethod;
import org.testng.annotations.BeforeMethod;
import org.testng.annotations.Test;

import java.util.Optional;

import static org.mockito.ArgumentMatchers.any;
import static org.mockito.ArgumentMatchers.anyLong;
import static org.mockito.Mockito.*;
import static org.testng.Assert.*;

public class CampaignServiceCreateHappyTest {

    @Mock private CampaignRepository campaignRepository;
    @Mock private CampaignTestCaseRepository campaignTestCaseRepository;
    @Mock private ProjectRepository projectRepository;
    @Mock private EnvironmentRepository environmentRepository;
    @Mock private TestCaseRepository testCaseRepository;
    @Mock private ProjectAccessService projectAccessService;

    @InjectMocks private CampaignService campaignService;

    private AutoCloseable mocks;
    private MockedStatic<SecurityUtils> mockedSecurity;

    @BeforeMethod
    public void setUp() {
        mocks = MockitoAnnotations.openMocks(this);
        mockedSecurity = mockStatic(SecurityUtils.class);
        mockedSecurity.when(SecurityUtils::getCurrentUserId).thenReturn(1L);
    }

    @AfterMethod
    public void tearDown() throws Exception {
        mockedSecurity.close();
        mocks.close();
    }

    @Test
    public void test_create_happy_path() {
        // Given
        Long projectId = 1L;
        Long environmentId = 5L;
        Long campaignId = 100L;
```

```
CreateCampaignRequest request = new CreateCampaignRequest();
request.setName("Release v2.3.0 Deployment");
request.setAppVersion("2.3.0");
request.setGitBranch("release/v2.3.0");
request.setEnvironmentId(environmentId);
request.setTriggerMode("SCHEDULED");
request.setTestCaseIds(java.util.Collections.emptyList());

Project project = new Project();
project.setId(projectId);
project.setGitDefaultBranch("main");

Environment environment = new Environment();
environment.setId(environmentId);
environment.setProject(project);

Campaign savedCampaign = new Campaign();
savedCampaign.setId(campaignId);
savedCampaign.setProject(project);
savedCampaign.setEnvironment(environment);
savedCampaign.setName("Release v2.3.0 Deployment");
savedCampaign.setAppVersion("2.3.0");
savedCampaign.setGitBranch("release/v2.3.0");
savedCampaign.setTriggerMode(Campaign.TriggerMode.SCHEDULED);
savedCampaign.setStatus(Campaign.CampaignStatus.DRAFT);
savedCampaign.setCreatedAt(java.time.LocalDateTime.now());

when(projectRepository.findById(projectId)).thenReturn(Optional.of(project));
doNothing().when(projectAccessService).checkMembership(project);
when(environmentRepository.findById(environmentId)).thenReturn(Optional.of(environment));
when(campaignRepository.save(any(Campaign.class))).thenReturn(savedCampaign);

// When
CampaignResponse response = campaignService.create(projectId, request);

// Then
assertNotNull(response);
assertEquals(response.getId(), campaignId);
assertEquals(response.getProjectId(), projectId);
assertEquals(response.getEnvironmentId(), environmentId);
assertEquals(response.getName(), "Release v2.3.0 Deployment");
assertEquals(response.getAppVersion(), "2.3.0");
assertEquals(response.getGitBranch(), "release/v2.3.0");
assertEquals(response.getTriggerMode(), "SCHEDULED");
assertEquals(response.getStatus(), "DRAFT");
assertNotNull(response.getCreatedAt());

verify(projectRepository).findById(projectId);
verify(projectAccessService).checkMembership(project);
verify(environmentRepository).findById(environmentId);
verify(campaignRepository).save(any(Campaign.class));
verifyNoInteractions(testCaseRepository, campaignTestCaseRepository);
}
}
```